WORKSHEET 1.2

Reading Limits off a Graph

Unit 1: Limits and Continuity

Key ideas

- To read a limit, cover the point and follow the curve toward it from each side.
- A filled dot gives the **value** $f(c)$; it says nothing about the limit.
- A limit fails to exist at a jump, where the function is unbounded, or where it oscillates.
- For $\lim_{x\to c} f(g(x))$: find what $g(x)$ approaches *and from which side*, then take that one-sided limit of f.

Worked examples

Example 1 · Limit and value at the same point

The graph of p is shown on the next page. Find $\lim_{x\to -1} p(x)$ and $p(-1)$.

Solution. From both sides the graph heads for the open circle at height 3. The filled dot at $x=-1$ is at height 1.

$\lim_{x\to -1} p(x) = 3$ **and** $p(-1) = 1$**.**

Example 2 · A jump

For the same function, find $\lim_{x\to 2^-} p(x)$, $\lim_{x\to 2^+} p(x)$ and $\lim_{x\to 2} p(x)$.

Solution. From the left the parabola comes down to the filled dot at height 0. From the right the line starts at the open circle at height 1.

$\lim_{x\to 2^-} p(x) = 0$**,** $\lim_{x\to 2^+} p(x) = 1$**, and** $\lim_{x\to 2} p(x)$ **does not exist.**

Example 3 · A composition

For the same function, find $\lim_{x\to 2^-} p(p(x))$.

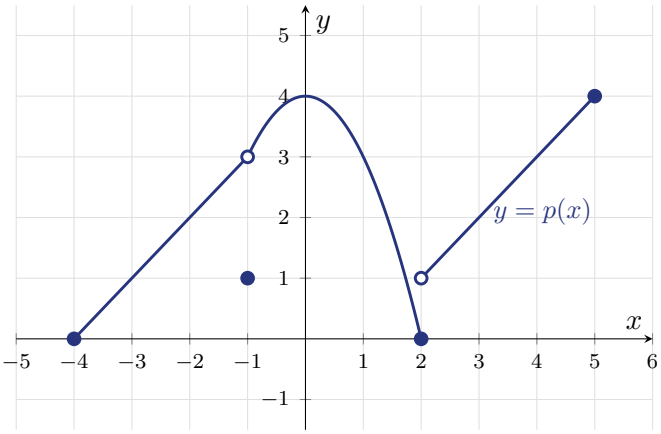
Solution. As $x\to 2^-$, $p(x) = 4 - x^2 \to 0$ through values *greater* than 0. So we need $\lim_{u\to 0^+} p(u) = 4$.

$\lim_{x\to 2^-} p(p(x)) = 4$**.**

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Practice

All questions on this page refer to the graph of p below, defined on $[-4, 5]$.



CORE

1. Find $\lim_{x \to 0} p(x)$.

2. Find $p(2)$.

3. Find $\lim_{x \to -1^+} p(x)$.

4. Find $\lim_{x \to 5^-} p(x)$.

EXAM LEVEL

5. Evaluate $\lim_{x \to -1} p(x) + p(2)$.

6. Evaluate $\lim_{x \to 1} [p(x)]^2$.

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7. For which value(s) of a in $(-4, 5)$ does $\lim_{x \to a} p(x)$ exist but differ from $p(a)$?

8. At how many values of x in $(-4, 5)$ is p discontinuous? List them.

Question 9 · Free-response practice

Use the graph of p on the previous page.

(a) Does $\lim_{x \to 2} p(x)$ exist? Justify your answer.

(b) Find $\lim_{x \to 2^+} p(p(x))$. Show your reasoning.

(c) Does $\lim_{x \to 2} p(p(x))$ exist? Justify, using part (b) and Example 3.

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Worked solutions

1. 4.
2. 0 (the filled dot).
3. 3.
4. 4.
5. $3 + 0 = 3$.
6. $p(x) \to 3$ as $x \to 1$, so the limit is 9.
7. $a = -1$ only: the limit is 3 but $p(-1) = 1$. (At $x = 2$ the limit does not exist.)
8. Two: $x = -1$ (limit 3, value 1) and $x = 2$ (a jump).
9. (a) No. $\lim_{x \to 2^-} p(x) = 0$ and $\lim_{x \to 2^+} p(x) = 1$. Since the one-sided limits are not equal, the limit does not exist.

   (b) As $x \to 2^+$, $p(x) = x - 1 \to 1$ through values greater than 1. Near $u = 1$ the graph is the parabola, which is continuous there, so the limit is $p(1) = 4 - 1 = 3$.

   (c) No. $\lim_{x \to 2^-} p(p(x)) = 4$ (Example 3) and $\lim_{x \to 2^+} p(p(x)) = 3$. They are not equal, so the limit does not exist.

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